Learning Each Person's Comfort Zone

Personalised greeting with reinforcement learning, on the Yanshee robot

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Seminar project demo

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The question

People differ in how close they want a robot to come and how loudly they want to be greeted. A robot can learn these preferences from how people react.

Does it need a different behaviour for each person, or is one well-trained behaviour enough for everyone?

The papers we reviewed (Wang et al. 2025, Tozadore et al. 2018, Gonzalez-Santocildes et al. 2024, Zedda et al. 2023) show that robots can adapt, but rarely compare against a trained single policy.

What the robot sees, does, and learns from

State (12 situations)

- distance to the person: intimate, personal, social, far
- its own greeting strength: low, medium, high

Actions (5)

- step closer, step back
- greet more strongly, more softly
- hold

Reward, from the person's reactions only

- touch on the robot: $+1$
- the person leaves: -2, interaction ends
- every step: -0.05

Learning: tabular Q-learning, a 12×5 table per person.

Three simulated people with hidden preferences: *reserved* (far, quiet), *neutral* (personal distance, medium), *sociable* (close, loud).

One policy per person, or one for everyone

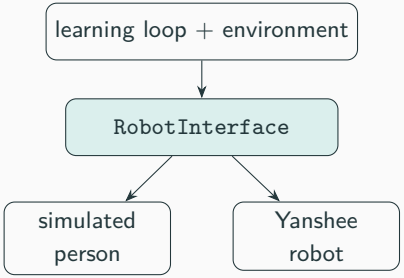
	reserved	neutral	sociable
scripted greeting	−1.26	2.03	−0.75
one policy for everyone	6.45	−0.75	−0.75
best single policy	11.81	−0.75	−0.75
one policy per person	**11.83**	**10.31**	**8.34**

Mean reward per interaction, 8 seeds, 6000 training episodes each.

The best single policy suits the reserved user and does worse than the scripted greeting for the other two. Learning a separate policy per person gains +11.06 and +9.09 on those two.

Pre-training on other people did not help a new person learn faster. Here the preference is the whole task, so there is nothing general to transfer.

Moving the same code to the Yanshee



The learning code is unchanged; only the robot class is new.

interface	on Yanshee
step closer / back	walk forward / backward
greeting strength	volume, gesture, spoken greeting
distance	infrared sensor
touch (reward)	touch sensor
person left	infrared sees nobody

Yanshee is 37 cm tall, so the distance zones are scaled by 1/3 onto a table: 15, 30 and 50 cm.

Policies are learned in simulation (about one second each) and run on the robot without further learning.

The live demo

A person plays one of the simulated users, following its rules.

1. **Reserved policy, reserved person.** The robot stops at a distance and greets quietly.
2. **Sociable policy, sociable person.** The robot walks up close and greets loudly.
3. **Sociable policy, reserved person.** The robot comes too close and the person leaves.
   This is the result in the table, in one interaction.

Limitations

- The users are simulated, and in the demo a person plays one. This shows the behaviour transfers to a robot, not how real people react.
- Tabular Q-learning on 12 states: easy to inspect, trains in a second, but too small for richer sensing.
- Three user types, no user study, no subjective measures.
- The robot is told which person it is greeting; recognising them is future work.

Code, results and documents: `github.com/nourgaser/nao-adaptive-rl`